MILITARY SPECIFICATION

TEST PLUGS, ELECTRICAL, FOR XM75E1

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for two types of test plugs entitled Test Plugs, Electrical which are used for completing circuits in the Nike Hercules warhead section.

1.2 Classification. This specification covers test plugs of the following:

Type I: Plug, Electrical, Test, XM75E1
Type II: Self Test Plug, 8822202

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connectors and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8822163	- Plug, Electrical, Test, XM75E1, Adaption Kit.
8822202	- Self Test Plug.
EL-8828999	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test plugs shall comply with all requirements specified on the drawings (dwgs) 8822163 and 8822202, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Continuity. The plugs shall exhibit point-to-point continuity in accordance with Table IV when tested as specified in 4.4.1.

3.5 Insulation resistance. The insulation resistance between all electrically isolated pins and the connector shell shall be not less than 20 megohms when tested as specified in 4.4.2.

3.6 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460(MU).

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwgs 8822163 and 8822202) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity (see 3.4)		X
3. Insulation resistance (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cap, electrical plug (see dwg 8822164 covering details of dwg 8822163).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of thread, max	Gage
	102. Minor diameter of thread, max	Gage
	103. Depth of threaded cavity, min	Gage
	104. Marking missing, incorrect, or unidentifiable	Visual
Minor:	None defined.	

4.3.2.1.2 Plug, electrical test, prior to assembling (see dwg 8822163).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Connecting wire loop missing or mislocated	Visual
Major:		
	101. Length of wire loops over max	Gage
	102. Improper soldering (see 3.6)	Visual
	103. Cracked plug body	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.7)	Visual

4.3.2.1.3 Plug, electrical test (see dwg 8822163).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length of ribbon, min	Gage
	102. Ribbon damaged or color incorrect	Visual
	103. Any component loose or missing	Visual
	104. Connector pins bent or damaged	Visual
	105. Insulation cracked or damaged	Visual

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*106.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
107.	Objective evidence relative to material missing or improper (see 3.1)	Visual

Minor:

201.	Gap between cap and connector over max	Gage
202.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.1.4 Cap, housing (see dwg 8822200 covering a detail of dwg 8822202).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Crack	Visual
102.	Nut plate missing	Visual
Minor:		
201.	Marking incorrect, missing, or unidentifiable	Visual

4.3.2.1.5 Self test plug assembly, prior to assembling housing cap (see dwg 8822202).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Jumper wire missing or mislocated	Visual
102.	Rachet action improper	Visual
103.	Any component missing or damaged	Visual
104.	Improper soldering (see 3.6)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.2.1.6 Self test plug assembly (see dwg 8822202).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any operation missing	Visual
102.	Any component missing, loose or damaged	Visual/Manual
103.	Connector pins bent or damaged	Visual
104.	Insulation cracked or damaged	Visual
105.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.1.7 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing in accordance with MIL-STD-1169	Visual
102.	Marking in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements.

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity and insulation resistance. Each test plug shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.4 and 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions specified on the Equipment Lists (EL) (See EL-8828999). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity. Using a suitable continuity indicating device, point to point continuity shall be determined in accordance with Table IV.

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Table IV

Type 1 (see dwg 8822163)

From	To
Pin A	Pins F, E and J
Pin B	Pin G
Pin P	Pin R
Pin O	Pin V
Pin W	Pins X, d and c
Pin D	Pins Z, Y and j
Pin N	Pin T
Pin I	Pin f
Pin M	Pin S
Pin K	Pins L, g and K
Pin U	Pin b
Pin a	Pins e and h

Type II (see dwg 8822202)

From	To
Pin A-1	Pins A-3, A-8, A-19, A-20, A-31, A-40, A-57, A-63, A-108, A-111, B-19, B-26, B-33, B-49, B-52, B-53, B-54, B-57, B-59, B-62, B-71, B-74, B-82, B-88, B-92, B-95 and B-96.
Pin A-2	Pins A-14, A-33, A-35, A-43, A-48, A-73, A-75, A-90, A-106, A-107, A-109, A-110, B-47, B-48, B-50, B-69 and B-89.
Pin A-55	Pins A-67, B-7, B-9, B-21, B-38, B-104, B-106, B-107, B-109 and B-110.

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Pin A-61	Pins A-81, B-11 and B-14.
Pin A-18	Pins A-45, A-80, B-16 and B-20.
Pin A-22	Pins A-58, B-5, B-75 and B-83.
Pin A-62	Pin A-78
Pin A-34	Pins B-34 and B-36.
Pin A-28	Pins A-29, A-52, B-78 and B-81.
Pin A-68	Pins B-32 and B-60.
Pin B-99	Pin B-111.
Pin B-30	Pins B-58 and B-61.
Pin B-28	Pins B-29 and B-68.
Pin A-23	Pins A-32, A-36, A-41, A-49, A-77, A-83, A-87, A-98, B-13 and B-76.
Pin A-26	Pins A-105, B-45, B-55, B-91, B-93, B-97 and B-98.
Pin A-9	Pins A-12, A-85, A-102, B-28 and B-73.
Pin A-82	Pin A-92.
Pin A-71	Pins B-35, B-51, B-90, B-100, B-101 and B-103.
Pin A-94	Pins A-99, A-100, A-86 and B-66.

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Pin A-39	Pins B-27 and B-46.
Pin A-88	Pins A-95 and B-64.
Pin A-104	Pins B-56 and B-63.
Pin A-65	Pins A-79, A-84, B-40 and B-72.
Pin A-93	Pin B-85.
Pin A-56	Pin B-6.
Pin A-69	Pins B-1 and B-86.
Pin A-24	Pins A-30, A-59, B-4 and B-14.
Pin A-54	Pin A-96.
Pin A-10	Pins A-11, A-17, B-39 and B-43.
Pin A-42	Pins A-50, B-2, B-37 and B-67.
Pin A-66	Pins A-74, B-108 and B-112.
Pin A-37	Pins A-44, B-102 and B-105.
Pin A-21	Pin B-17.

4.4.2 Insulation resistance (see 3.5). The insulation resistance shall be measured with approved insulation resistance measuring equipment using a potential of 500 ± 50 volts direct current (vdc) with the current limited to 5 milliamperes (ma).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test plugs shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The test plugs shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

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5.2 Packing.

5.2.1 Levels A and C. The test plugs shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1798 Revision 2 dated 25 July 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A155

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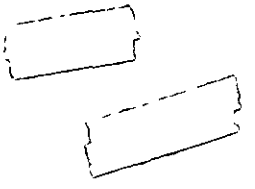
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1426
1 JAN 72

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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